Mechanics & Astronomy **65** 274206

100. Fromme T, Reichenberger S, Tibbetts K M, and Barcikowski S 2024 Laser synthesis of nanoparticles in organic solvents – products, reactions, and perspectives *Beilstein Journal of Nanotechnology* **15** 638-663
101. De Giacomo A, Gaudiuso R, Dell'Aglio M, and Santagata A 2010 The role of continuum radiation in laser induced plasma spectroscopy *Spectrochimica Acta Part B: Atomic Spectroscopy* **65** 385-394
102. Yang Z, Zhang C, Ying K, Li Q, Zhang H, Zhang H, and Lu J 2024 Transient optical properties change and dense plasma dynamics during water breakdown induced by ultrashort laser pulses *Phys. Plasmas* **31** 050703
103. Tian Y, et al. 2025 Temperature determination of laser-induced plasma in water *Plasma Sources Science and Technology* **34** 025012
104. Vogel A, Busch S, Jungnickel K, and Birngruber R 1994 Mechanisms of intraocular photodisruption with picosecond and nanosecond laser-pulses *Laser Surg. Med.* **15** 32-43
105. Freidank S, Vogel A, and Linz N 2022 Mechanisms of corneal intrastromal laser dissection for refractive surgery: ultra-high-speed photographic investigation at up to 50 million frames per second *Biomed. Opt. Express* **13** 3056-3079
106. Zysset B, Fujimoto J G, and Deutsch T F 1989 Time-Resolved Measurements of Picosecond Optical-Breakdown *Appl Phys B-Photo* **48** 139-147
107. Vogel A, Capon M R C, Asiyo-Vogel M N, and Birngruber R 1994 Intraocular Photodisruption with Picosecond and Nanosecond Laser-Pulses - Tissue Effects in Cornea, Lens, and Retina *Invest Ophth Vis Sci* **35** 3032-3044
108. Vogel A, Nahen K, Theisen D, and Noack J 1996 Plasma formation in water by picosecond and nanosecond Nd: YAG laser pulses - Part I: Optical breakdown at threshold and superthreshold irradiance. *IEEE J. Sel. Topics Quantum Electron.* **2** 847-860
109. Sacchi C A 1991 Laser-induced electric breakdown in water *J. Opt. Soc. B* **8** 337-345
110. Noack J, Hammer D X, Noojin G D, Rockwell B A, and Vogel A 1998 Influence of pulse duration on mechanical effects after laser-induced breakdown in water *J. Appl. Phys.* **83** 7488-7495
111. Juhasz T, Hu X H, Turi L, and Bor Z 1994 Dynamics of Shock-Waves and Cavitation Bubbles Generated by Picosecond Laser-Pulses in Corneal Tissue and Water *Laser Surg. Med.* **15** 91-98
112. Loesel F H, Niemz M H, Bille J F, and Juhasz T 1996 Laser-induced optical breakdown on hard and soft tissues and its dependence on the pulse duration: Experiment and model *IEEE J. Quantum Electron.* **32** 1717-1722
113. Kennedy P K 1995 A first-order model for computation of laser-induced breakdown thresholds in ocular and aqueous media: Part 1 - Theory *IEEE J. Quantum Electron.* **31** 2241-2249
114. Vogel A, et al. 1999 Energy balance of optical breakdown in water at nanosecond to femtosecond time scales *Appl. Phys. B.* **68** 271-280
115. Noack J and Vogel A 1999 Laser-induced plasma formation in water at nanosecond to femtosecond time scales: calculation of thresholds, absorption coefficients, and energy density *IEEE J. Quantum Electron.* **35** 1156-1167
116. Konig K, Riemann I, and Fritzsche W 2001 Nanodissection of human chromosomes with near-infrared femtosecond laser pulses *Opt. Lett.* **26** 819-821
117. Vogel A, Linz N, Freidank S, and Paltauf G 2008 Femtosecond laser induced nanocavitation in water:

Accepted Manuscript

implications for optical breakdown threshold and cell surgery *Phys. Rev. Lett.* **100** 038102

118. Heisterkamp A, Maxwell I Z, Mazur E, Underwood J M, Nickerson J A, Kumar S, and Ingber D E 2005 Pulse energy dependence of subcellular dissection by femtosecond laser pulses *Opt. Express* **13** 3690-3696
119. Davis A A, Farrar M J, Nishimura N, Jin M M, and Schaffer C B 2013 Optoporation and Genetic Manipulation of Cells Using Femtosecond Laser Pulses *Biophys. J.* **105** 862-871
120. Glezer E N and Mazur E 1997 Ultrafast-laser driven micro-explosions in transparent materials *Appl. Phys. Lett.* **71** 882-884
121. Juodkazis S, Nishimura K, Tanaka S, Misawa H, Gamaly E G, Luther-Davies B, Hallo L, Nicolai P, and Tikhonchuk V T 2006 Laser-induced microexplosion confined in the bulk of a sapphire crystal: Evidence of multimegabar pressures *Phys. Rev. Lett.* **96** 166101
122. Gamaly E G, Juodkazis S, Nishimura K, Misawa H, and Luther-Davies B 2006 Laser-matter interaction in the bulk of a transparent solid: Confined microexplosion and void formation *Phys. Rev. B* **73** 214101
123. Schaffer C, Nishimura N, Glezer E, Kim A, and Mazur E 2002 Dynamics of femtosecond laser-induced breakdown in water from femtoseconds to microseconds *Opt. Express* **10** 196-203
124. Hutson M and Ma X 2007 Plasma and cavitation dynamics during pulsed laser microsurgery in vivo *Phys. Rev. Lett.* **99** 158104
125. Stuart B C, Feit M D, Herman S, Rubenchik A M, Shore B W, and Perry M D 1996 Nanosecond-to-femtosecond laser-induced breakdown in dielectrics *Phys. Rev. B* **53** 1749-1761
126. Lenzner M, Kruger J, Sartania S, Cheng Z, Spielmann C, Mourou G, Kautek W, and Krausz F 1998 Femtosecond optical breakdown in dielectrics *Phys. Rev. Lett.* **80** 4076-4079
127. Liu X, Du D, and Mourou G 1997 Laser ablation and micromachining with ultrashort laser pulses *IEEE J. Quantum Electron.* **33** 1706-1716
128. Joglekar A P, Liu H, Spooner G J, Meyhofer E, Mourou G, and Hunt A J 2003 A study of the deterministic character of optical damage by femtosecond laser pulses and applications to nanomachining *Appl. Phys. B* **77** 25-30
129. Bass M and Barrett H H 1972 Avalanche breakdown and the probabilistic nature of laser-induced damage *IEEE J. Quantum Electron.* **8** 338-343
130. Arnold C L, Heisterkamp A, Ertmer W, and Lubatschowski H 2007 Computational model for nonlinear plasma formation in high NA micromachining of transparent materials and biological cells *Opt. Express* **15** 10303-10317
131. Fedorov V Y, Chanal M, Grojo D, and Tzortzakis S 2016 Accessing Extreme Spatiotemporal Localization of High-Power Laser Radiation through Transformation Optics and Scalar Wave Equations *Phys. Rev. Lett.* **117** 043902
132. Raizer Y P 1966 Breakdown and heating of gases under the influence of a laser beam *Sov. Phys. Usp.* **8** 650-673
133. Docchio F, Regondi P, Capon M R C, and Mellerio J 1988 Study of the temporal and spatial dynamics of plasmas induced in liquids by nanosecond Nd-YAG laser-pulses .1. Analysis of the plasma starting times *Appl. Opt.* **27** 3661-3668
134. Docchio F, Regondi P, Capon M R C, and Mellerio J 1988 Study of the temporal and spatial dynamics of plasmas induced in liquids by nanosecond Nd YAG laser-pulses .2. Plasma luminescence and shielding *Appl. Opt.* **27** 3669-3674

Accepted Manuscript

135. Hammer D X, Jansen E D, Frenz M, Noojin G D, Thomas R J, Noack J, Vogel A, Rockwell B A, and Welch A J 1997 Shielding properties of laser-induced breakdown in water for pulse durations from 5 ns to 125 fs *Appl. Opt.* **36** 5630-5640

136. Bulgakova N M, Zhukov V P, Sonina S V, and Meshcheryakov Y P 2015 Modification of transparent materials with ultrashort laser pulses: What is energetically and mechanically meaningful? *J. Appl. Phys.* **118** 233108

137. Perry M D, Stuart B C, Banks P S, Feit M D, Yanovsky V, and Rubenchik A M 1999 Ultrashort-pulse laser machining of dielectric materials *J. Appl. Phys.* **85** 6803-6810

138. Theobald W, Hassner R, Kingham R, Sauerbrey R, Fehr R, Gericke D O, Schlanges M, Kraeft W D, and Ishikawa K 1999 Electron densities, temperatures, and the dielectric function of femtosecond-laser-produced plasmas *Phys. Rev. E* **59** 3544-3553

139. Feit M D, Komashko A M, and Rubenchik A M 2004 Ultra-short pulse laser interaction with transparent dielectrics *Appl. Phys. A* **79** 1657-1661

140. Couairon A and Mysyrowicz A 2007 Femtosecond filamentation in transparent media *Phys. Rep.* **441** 47-189

141. Couairon A, Brambilla E, Corti T, Majus D, Ramírez-Góngora O D, and Kolesik M 2011 Practitioner's guide to laser pulse propagation models and simulation *Eur Phys J-Spec Top* **199** 5-76

142. Kaiser A, Rethfeld B, Vicanek M, and Simon G 2000 Microscopic processes in dielectrics under irradiation by subpicosecond laser pulses *Phys. Rev. B* **61** 11437-11450

143. Christensen B H and Balling P 2009 Modeling ultrashort-pulse laser ablation of dielectric materials *Phys. Rev. B* **79** 155424

144. Keldysh L V 1965 Ionization in the field of a strong electromagnetic wave *Sov. Phys. Jetp.* **20** 1307-1314

145. Ammosov M V, Delone N B, and Krainov V P 1986 Tunnel ionization of complex atoms and atomic ions in a varying electromagnetic-field *Zh. Eksp. Teor. Fiz.* **91** 2008-2013

146. Rethfeld B 2004 Unified model for the free-electron avalanche in laser-irradiated dielectrics *Phys. Rev. Lett.* **92** 187401

147. Grojo D, Leyder S, Delaporte P, Marine W, Sentis M, and Uteza O 2013 Long-wavelength multiphoton ionization inside band-gap solids *Phys. Rev. B* **88** 195135

148. Sarpe C, Köhler J, Winkler T, Wollenhaupt M, and Baumert T 2012 Real-time observation of transient electron density in water irradiated with tailored femtosecond laser pulses *New J. Phys.* **14** 075021

149. Liang X X, Zhang Z X, and Vogel A 2019 Multi-rate-equation modeling of the energy spectrum of laser-induced conduction band electrons in water *Opt. Express* **27** 4672-4693

150. Rethfeld B 2006 Free-electron generation in laser-irradiated dielectrics *Phys. Rev. B* **73** 035101

151. Rethfeld B, Brenk O, Medvedev N, Krutsch H, and Hoffmann D H H 2010 Interaction of dielectrics with femtosecond laser pulses: Application of kinetic approach and multiple rate equation *Appl. Phys. A* **101** 19-25

152. Caruso F and Novko D 2022 Ultrafast dynamics of electrons and phonons: from the two-temperature model to the time-dependent Boltzmann equation *Advances in Physics: X* **7** 2095925

153. Rethfeld B, Ivanov D S, Garcia M E, and Anisimov S I 2017 Modelling ultrafast laser ablation *Journal of Physics D: Applied Physics* **50** 193001

154. Nikogosyan D N, Oraevsky A A, and Rupasov V I 1983 Two-photon ionization and dissociation of liquid water by powerful laser UV radiation *Chem. Phys.* **77** 131-143

Accepted Manuscript